

ETV6 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes ETV6 gene fusions from the msk_impact_50k_2026 study, covering 90 fusion events. 64/90 fusions (71.1%) were in-frame. 68/90 fusions (75.6%) retained the Sterile alpha motif (SAM)/Pointed domain. Domain retention was tested with Fisher's exact test ($p=5.12326e-06$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for ETV6.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 90 analyzed fusion events, identifying 27 distinct fusion partner genes. The most frequent partner was NTRK3, observed in 55 events.

Domain retention

Sterile alpha motif (SAM)/Pointed domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 58/64 (90.6%) of in-frame fusions retained the domain, $p=5.12326e-06$ (statistically significant at $\alpha=0.05$). The Sterile alpha motif (SAM)/Pointed domain appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

Domain disruption

Domain-disruption analysis was skipped: ETV6 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 88 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 155 aa (permutation-corrected $p=0.168317$, not statistically significant at $\alpha=0.05$). This is 32 aa from the nearest configured domain boundary at 123 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=68$) and "not_retained" ($n=20$). For Frame_status, retained: 58/66 (87.9%, 95% CI 77.9%-93.7%); not_retained: 6/13 (46.2%, 95% CI 23.2%-70.9%). Welch's t-test comparing Tumor_variant_count between groups gave means of 73.6765 vs 31.15, $p=0.00082599$ (statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: ETV6 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
NTRK3	55	0.6111111111111112	0.20043213737826426	None	0.3840220385674932	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.44910851064190527	1
ABCC9	1	0.011111111111111112	0.20043213737826426	None	1.0	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3396043784105003	2
ANKS1B	1	0.011111111111111112	0.20043213737826426	None	1.0	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3396043784105003	3
ARHGAP26	1	0.011111111111111112	0.20043213737826426	None	1.0	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3396043784105003	4
CUX2	1	0.011111111111111112	0.20043213737826426	None	1.0	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3396043784105003	5
GALNT18	1	0.011111111111111112	0.20043213737826426	None	1.0	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3396043784105003	6
GNB1	1	0.011111111111111112	0.20043213737826426	None	1.0	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3396043784105003	7
GRIN2B	1	0.011111111111111112	0.20043213737826426	None	1.0	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3396043784105003	8
NOL4	1	0.011111111111111112	0.20043213737826426	None	1.0	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3396043784105003	9
PLEKHG7	1	0.011111111111111112	0.20043213737826426	None	1.0	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3396043784105003	10
PTPRN2	1	0.011111111111111112	0.20043213737826426	None	1.0	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3396043784105003	11

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
BORCS5	3	0.0333333333333333	0.20043213737826426	None	0.4747474747474748	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.24945286325898514	12
HCFC1	2	0.0222222222222222	0.20043213737826426	None	0.38095238095238093	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.22769961650573836	13
CDKN1B	1	0.0111111111111111	0.20043213737826426	None	0.38095238095238093	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.22353294983907168	14
BCL2L14	3	0.0333333333333333	0.20043213737826426	None	0.212121212121215	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.2002104390165609	15
SGO2	2	0.0222222222222222	0.20043213737826426	None	0.212121212121215	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.19604377234989423	16
SLIT2	2	0.0222222222222222	0.20043213737826426	None	0.212121212121215	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.19604377234989423	17
CCND2	2	0.0222222222222222	0.20043213737826426	None	None	0.6824213505050083	['confidence_certainty', 'domain_retention', 'recurrence']	0.1923335939411286	18
AEBP2	1	0.0111111111111111	0.20043213737826426	None	0.212121212121215	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.19187710568322755	19
APOLD1	1	0.0111111111111111	0.20043213737826426	None	0.212121212121215	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.19187710568322755	20
FAM234B	1	0.0111111111111111	0.20043213737826426	None	0.212121212121215	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.19187710568322755	21
FGFR1OP2	1	0.0111111111111111	0.20043213737826426	None	0.212121212121215	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.19187710568322755	22
IKZF3	1	0.0111111111111111	0.20043213737826426	None	0.212121212121215	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.19187710568322755	23
LRP6	1	0.0111111111111111	0.20043213737826426	None	0.212121212121215	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.19187710568322755	24

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
PDE3A	1	0.011111111111111112	0.20043213737826426	None	0.212121212121215	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.19187710568322755	25
SUFU	1	0.011111111111111112	0.20043213737826426	None	0.212121212121215	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.19187710568322755	26
PPIL2	2	0.022222222222222223	0.20043213737826426	None	0.0	0.6824213505050083	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.15627104507716696	27

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
12	2	66	1	19	0.5757575757575758	0.5433039294306334	0.26495715393209934
155	17	51	9	11	0.4074074074074074	0.10005714167068805	0.9997519077526134
337	67	1	19	1	3.526315789473684	0.40491118077324983	0.3926402309345144
385	68	0	19	1	None	0.22727272727272727	0.6434526764861874

domain_retention tables

frame_domain_contingency_table

58	10
6	14

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	Sterile alpha motif (SAM)/Pointed domain	88	68	0	20	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
2	68	90	0.7555555555555555

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ABCC9	1
AEBP2	1
ANKS1B	1
APOLD1	1
ARHGAP26	1
BCL2L14	3
BORCS5	3
CCND2	2
CDKN1B	1
CUX2	1
FAM234B	1
FGFR1OP2	1
GALNT18	1
GNB1	1
GRIN2B	1
HCFC1	2
IKZF3	1
LRP6	1
NOL4	1
NTRK3	55
PDE3A	1
PLEKHG7	1
PPIL2	2
PTPRN2	1
SGO2	2
SLIT2	2
SUFU	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 5637-T01-I M6-3	P-0055 637-T0 1-IM6		ARHGAP26-ETV6	ARHGAP26	out-of-frame	three_prime	12013736	4	155	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		ARHGAP26 (NM_015071) - ETV6 (NM_001987) rearrangement: t(5;12)(q31.3;p13.2)(chr5:g.142309261::chr12:g.12013736) Protein Fusion: out of frame {ARHGAP26:ETV6}	NA
EVT-P-005 1300-T01-I M6-4	P-0051 300-T0 1-IM6		BCL2L14-ETV6	BCL2L14	in-frame	three_prime	12031156	6	337	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		BCL2L14 (NM_138723) - ETV6 (NM_001987) rearrangement: c.433+1623:BCL2L14_c.1010-6223:ETV6dup Protein Fusion: in frame {BCL2L14:ETV6}	NA
EVT-P-004 0855-T01-I M6-5	P-0040 855-T0 1-IM6		BCL2L14-ETV6	BCL2L14	in-frame	three_prime	12027991	5	337	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		BCL2L14 (NM_138723) - ETV6 (NM_001987) rearrangement: c.434-2555:BCL2L14_c.1009+5088:ETV6dup Protein Fusion: in frame {BCL2L14:ETV6}	NA
EVT-P-001 5724-T01-I M6-6	P-0015 724-T0 1-IM6		BORCS5-ETV6	BORCS5	in-frame	three_prime	12034428	6	337	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		LOH12CR1 (NM_058169) - ETV6 (NM_001987) rearrangement: c.202+1037:LOH12CR1_c.1010-2951:ETV6dup Protein fusion: in frame (LOH12CR1-ETV6)	NA
EVT-P-003 8806-T01-I M6-11	P-0038 806-T0 1-IM6		CUX2-ETV6	CUX2	out-of-frame	three_prime	12008261	4	155	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		CUX2 (NM_015267) - ETV6 (NM_001987) Rearrangement: c.63+46462:CUX2_c.463+1766:ETV6 Protein Fusion: out of frame {CUX2:ETV6}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0007927-T01-IM5-12	P-0007927-T01-IM5		ETV6-ABCC9	ABCC9	out-of-frame	five_prime	12015620	5	155	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - ABCC9 (NM_020297) rearrangement: c.464-6738:ETV6_c.3473-739:ABCC9inv Protein fusion: out of frame (ETV6-ABCC9)	NA
EVT-P-0060870-T01-IM7-13	P-0060870-T01-IM7		ETV6-AEBP2	AEBP2	out-of-frame	five_prime	12025073	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - AEBP2 (NM_00114176) rearrangement: c.1009+2170:ETV6_c.880-2512:AEBP2del Protein Fusion: out of frame {ETV6:AEBP2}	NA
EVT-P-0019836-T01-IM6-14	P-0019836-T01-IM6		ETV6-ANKS1B	ANKS1B	in-frame	five_prime	12006878	4	155	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - ANKS1B (NM_152788) Rearrangement : c.463+383:ETV6_c.1757-48649:ANKS1Binv Protein Fusion: in frame {ETV6:ANKS1B}	NA
EVT-P-0027889-T01-IM6-15	P-0027889-T01-IM6		ETV6-APOLD1	APOLD1	out-of-frame	five_prime	12030422	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6:NM_001987) - APOLD1 (NM_001130415) rearrangement: c.1010-6957:ETV6_c.96+30120:APOLD1del Protein Fusion: out of frame {ETV6:APOLD1}	NA
EVT-P-0025013-T01-IM6-19	P-0025013-T01-IM6		ETV6-BCL2L14	BCL2L14	out-of-frame	three_prime	12025353	5	337	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		BCL2L14 (NM_138722) -ETV6 (NM_001987) Rearrangement : c.945+502:BCL2L14_c.1009+2450:ETV6dup Protein Fusion: out of frame {BCL2L14:ETV6}	NA
EVT-P-0009946-T01-IM5-22	P-0009946-T01-IM5		ETV6-BORCS5	BORCS5	out-of-frame	three_prime	12034181	6	337	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		LOH12CR1 (NM_058169) - ETV6 (NM_001987) rearrangement: c.361-7873:LOH12CR1_c.1010-3198:ETV6dup Protein fusion: out of frame (LOH12CR1-ETV6)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 3635-T01-I M6-23	P-0033 635-T0 1-IM6		ETV6- BORCS5	BORCS5	in-frame	three_prime	12019982	5	155	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		LOH12CR1 (NM_058169) - ETV6 (NM_001987) rearrangement: c.203-15769: LOH12CR1_c.464-2376:ETV6dup Protein Fusion: in frame {LOH12CR1:ETV6}	NA
EVT-P-001 1506-T01-I M5-24	P-0011 506-T0 1-IM5		ETV6- CDKN1B	CDKN1B	out-of-frame	five_prime	11894784	2	12	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		ETV6 (NM_001987) - CDKN1B (NM_004064) rearrangement: c.34-10600:ETV6_c.476-58:CDKN1Bdel Protein fusion: out of frame (ETV6-CDKN1B)	NA
EVT-P-002 2579-T01-I M6-121	P-0022 579-T0 1-IM6		ETV6- FAM234B	FAM234B	in-frame	three_prime	12035084	6	337	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		KIAA1467 (NM_020853) - ETV6 (NM_001987) rearrangement: c.1643-1210: KIAA1467_c.1010-2295:ETV6dup Protein Fusion: in frame {KIAA1467:ETV6}	NA
EVT-P-000 7927-T01-I M5-122	P-0007 927-T0 1-IM5		ETV6- FGFR1OP2	FGFR1OP2	unknown	three_prime	12029024	5	337	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		FGFR1OP2 (NM_015633) - ETV6 (NM_001987) rearrangement: c.1009+6121: ETV6_c.-14-5021:FGFR1OP2dup Transcript fusion (FGFR1OP2-ETV6)	NA
EVT-P-002 8989-T01-I M6-123	P-0028 989-T0 1-IM6		ETV6- GALNT18	GALNT18	unknown	three_prime	12014281	4	155	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		ETV6 (NM_001987) rearrangement: t(11;12)(p15.3;p13.2)(chr11:g.11366394::chr12:g.12014281). Antisense Fusion	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 6430-T01-I M6-124	P-0026 430-T0 1-IM6		ETV6- GNB1	GNB1	unknown	five_prime	12019513	5	155	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - GNB1 (NM_002074) rearrangement: t(1;12)(p36/33;p13.2)(chr1:g.1785936::chr12:g.12019513) Transcript Fusion {ETV6:GNB1}	NA
EVT-P-001 9445-T01-I M6-125	P-0019 445-T0 1-IM6		ETV6- GRIN2B	GRIN2B	unknown	three_prime	12009253	4	155	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		ETV6 (NM_001987) - GRIN2B (NM_000834) rearrangement: c.463+2758:ETV6_c.-19+139 31:GRIN2Bdel Antisense Fusion	NA
EVT-P-002 3544-T01-I M6-133	P-0023 544-T0 1-IM6		ETV6- LRP6	LRP6	unknown	three_prime	12035485	6	337	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		ETV6 (NM_001987) rearrangement: c.1010-1894_chr12:g.12422377del Antisense Fusion	NA
EVT-P-003 3921-T02-I M6-135	P-0033 921-T0 2-IM6		ETV6- NTRK3	NTRK3	in-frame	five_prime	12031161	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 3921-T04-I M6-136	P-0033 921-T0 4-IM6		ETV6- NTRK3	NTRK3	in-frame	five_prime	12031161	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-006 1265-T01-I M7-137	P-0061 265-T0 1-IM7		ETV6- NTRK3	NTRK3	in-frame	five_prime	12033175	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12033175::chr15:g.8854122) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 7610-T01-I M6-138	P-0037 610-T0 1-IM6		ETV6- NTRK 3	NTR K3	in-frame	five_prime	120146 63	5	155	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) rearrangement: t(12;15)(p13.2;q25.3)(chr12:g.12014663::chr15:g.88594532) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 5767-T06-I M7-139	P-0035 767-T0 6-IM7		ETV6- NTRK 3	NTR K3	in-frame	five_prime	120291 43	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12029143::chr15:g.88505787) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 3921-T05-I M7-140	P-0033 921-T0 5-IM7		ETV6- NTRK 3	NTR K3	in-frame	five_prime	120311 61	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 5767-T04-I M6-141	P-0035 767-T0 4-IM6		ETV6- NTRK 3	NTR K3	in-frame	five_prime	120291 43	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12029143::chr15:g.88505787) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-004 0355-T01-I M6-142	P-0040 355-T0 1-IM6		ETV6- NTRK 3	NTR K3	in-frame	five_prime	120361 11	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12036111::chr15:g.88497191) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-001 3664-T02-I M5-143	P-0013 664-T0 2-IM5		ETV6- NTRK 3	NTR K3	in-frame	five_prime	120320 77	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) Fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q26.1)(chr12:g.12032077::chr15:g.88558191) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-004 2503-T01-I M6-144	P-0042 503-T0 1-IM6		ETV6- NTRK 3	NTR K3	in-frame	five_prime	120328 02	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12032802::chr15:g.88556067) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 5703-T01-I M6-145	P-0055 703-T0 1-IM6		ETV6- NTRK 3	NTR K3	in-frame	five_prime	120247 85	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12024785::chr15:g.88505980) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-006 4964-T01-I M7-146	P-0064 964-T0 1-IM7		ETV6- NTRK 3	NTR K3	in-frame	five_prime	120323 10	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12032310::chr15:g.88564682) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-005 6156-T02-I M7-147	P-0056 156-T0 2-IM7		ETV6- NTRK 3	NTR K3	in-frame	five_prime	120119 57	4	155	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12011957::chr15:g.88624274) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-001 3664-T04-I M6-148	P-0013 664-T0 4-IM6		ETV6- NTRK 3	NTR K3	in-frame	five_prime	120320 77	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032077::chr15:g.88558191) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-005 8408-T01-I M6-149	P-0058 408-T0 1-IM6		ETV6- NTRK 3	NTR K3	in-frame	five_prime	120357 60	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12035760::chr15:g.88492156) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 2148-T04-I M6-150	P-0032 148-T0 4-IM6		ETV6- NTRK 3	NTR K3	in-frame	five_prime	120372 17	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12037217::chr15:g.88522437) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 3810-T01-I M6-152	P-0033 810-T0 1-IM6		ETV6- PDE3 A	PDE 3A	out-of-frame	five_prime	120361 40	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) -PDE3A (NM_000921) Rearrangement : c.1010-1239:ETV6_c.960+53099:PDE3A Protein Fusion: out of frame {ETV6:PDE3A}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 0402-T01-I M6-154	P-0020 402-T0 1-IM6		ETV6- PLEK HG7	PLE KH G7	unknown	five_prime	120169 69	5	155	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) rearrangement: c.464-5389:E TV6_chr12:g.93097480del Transcript Fusion {ETV6:C12orf74}	NA
EVT-P-001 1048-T03-I M7-155	P-0011 048-T0 3-IM7		ETV6- PPIL2	PPI L2	in-frame	five_prime	120388 44	7	385	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - PPIL2 (NM_014337) rearrangement: t(12;22)(p13.2;q11.21)(chr12: g.12038844::chr22:g.2204188 5) Protein Fusion: in frame {ETV6:PPIL2}	NA
EVT-P-003 5512-T01-I M6-156	P-0035 512-T0 1-IM6		ETV6- PTPR N2	PTP RN2	unknown	three_prime	120129 58	4	155	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		ETV6 (NM_001987) rearrangement: t(7;12)(q36.3; p13.2)(chr7:g.158331279::chr 12:g.12012958) Antisense Fusion	NA
EVT-P-005 2787-T01-I M6-159	P-0052 787-T0 1-IM6		ETV6- SGO2	SG O2	out-of-frame	five_prime	120288 02	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - SGOL2 (NM_152524) rearrangement: t(2;12)(q33.1;p13.2)(chr2:g.20 1410903::chr12:g.12028802) Protein Fusion: out of frame {ETV6:SGOL2}	NA
EVT-P-005 2787-T03-I M6-160	P-0052 787-T0 3-IM6		ETV6- SGO2	SG O2	out-of-frame	five_prime	120288 03	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - SGOL2 (NM_152524) rearrangement: t(2;12)(q33.1;p13.2)(chr2:g.20 1410918::chr12:g.12028803) Protein Fusion: out of frame {ETV6:SGOL2}	NA
EVT-P-002 1918-T01-I M6-161	P-0021 918-T0 1-IM6		ETV6- SUFU	SUF U	in-frame	five_prime	120334 06	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - SUFU (NM_016169) rearrangement: t(10;12)(q24.32;p13.2)(chr10: g.104320248::chr12:g.120334 06) Protein Fusion: in frame {ETV6:SUFU}	NA
EVT-P-003 6181-T01-I M6-163	P-0036 181-T0 1-IM6		HCFC 1-ETV 6	HCF C1	out-of-frame	three_prime	119053 67	2	12	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			HCFC1 (NM_005334) - ETV6 (NM_0019870) rearrangement: t(X;12)(q28;p 13.2)(chrX:g.153221893::chr1 2:g.11905367) Protein Fusion: out of frame {HCFC1:ETV6}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 6181-T06-I M7-164	P-0036 181-T0 6-IM7		HCFC1-ETV6	HCFC1	out-of-frame	three_prime	11905373	2	12	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			HCFC1 (NM_005334) - ETV6 (NM_001987) rearrangement: t(12;X)(p13.2;q28)(chr12:g.11905373::chrX:g.153221893) Protein Fusion: out of frame {HCFC1:ETV6}	NA
EVT-P-002 0738-T01-I M6-165	P-0020 738-T0 1-IM6		IKZF3-ETV6	IKZF3	in-frame	five_prime	12023694	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - IKZF3(NM_012481) rearrangement: t(12;17)(p13.2;q12)(chr12:g.12023694::chr17:g.38012519) Protein Fusion: in frame {ETV6:IKZF3}	NA
EVT-P-003 6266-T01-I M6-166	P-0036 266-T0 1-IM6		NOL4-ETV6	NOL4	unknown	three_prime	12015004	5	155	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		ETV6 (NM_001987) - NOL4 (NM_003787) rearrangement: t(12;18)(p13.2;q12.1)(chr12:g.12015004::chr18:g.31639519) Antisense Fusion	NA
EVT-P-000 2978-T03-I M5-167	P-0002 978-T0 3-IM5		NTRK3-ETV6	NTRK3	in-frame	five_prime	12028576	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 5 fused to NTRK3 exons 15 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-001 6507-T02-I M6-168	P-0016 507-T0 2-IM6		NTRK3-ETV6	NTRK3	in-frame	five_prime	12023709	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-14 fused with NTRK3 exons 6-20 including the kinase domain): t(12,15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-000 2978-T06-I M6-169	P-0002 978-T0 6-IM6		NTRK3-ETV6	NTRK3	in-frame	five_prime	12028576	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 3196-T03-I M6-170	P-0013 196-T0 3-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120360 72	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072::chr15:g.88501357) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-000 2978-T05-I M6-171	P-0002 978-T0 5-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120285 76	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-000 2671-T01-I M3-172	P-0002 671-T0 1-IM3		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120358 13	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			NA Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-003 5767-T01-I M6-173	P-0035 767-T0 1-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120291 43	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12029143::chr15:g.88505787) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 5552-T01-I M6-174	P-0035 552-T0 1-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120209 63	5	155	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12020963::chr15:g.88651251) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-000 2978-T04-I M5-175	P-0002 978-T0 4-IM5		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120285 76	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007772-T02-I M5-176	P-0007772-T02-IM5		NTRK3-ETV6	NTRK3	in-frame	three_prime	12020258	5	155	True	lost	0.0	False		Sterile alpha motif (SAM)/Pointed domain		ETV6 (NM_001012338) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257::chr15:g.88600844) Protein fusion: in frame (NTRK3-ETV6)	NA
EVT-P-0013196-T02-I M6-177	P-0013196-T02-IM6		NTRK3-ETV6	NTRK3	in-frame	five_prime	12036072	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072::chr15:g.88501357) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0007772-T03-I M5-178	P-0007772-T03-IM5		NTRK3-ETV6	NTRK3	in-frame	five_prime	12020257	5	155	True	retained	1.0	False	Sterile alpha motif (SAM)/Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257::chr15:g.88600844) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0007911-T02-I M5-179	P-0007911-T02-IM5		NTRK3-ETV6	NTRK3	in-frame	five_prime	12007142	4	155	True	retained	1.0	False	Sterile alpha motif (SAM)/Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 4 fused with NTRK3 exons 13-20): t(12;15)(p13.2;q25.3)(chr12:g.12007142::chr15:g.88592430) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0010264-T01-I M5-180	P-0010264-T01-IM5		NTRK3-ETV6	NTRK3	in-frame	five_prime	12014962	5	155	True	retained	1.0	False	Sterile alpha motif (SAM)/Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused in frame with NTRK3 exons 13-20): t(12;15)(p13.2;q25.3)(chr12:g.12014962::chr15:g.88588255) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0027006-T04-I M6-181	P-0027006-T04-IM6		NTRK3-ETV6	NTRK3	in-frame	five_prime	12032352	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032352::chr15:g.88512343) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 7006-T03-I M6-182	P-0027 006-T0 3-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120323 52	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12037037::chr15:g.88514531) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-002 7006-T01-I M6-183	P-0027 006-T0 1-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120323 52	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12032352::chr15:g.88512343) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-002 6696-T01-I M6-184	P-0026 696-T0 1-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120079 67	4	155	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20):t(12;15)(p13.2;q25.3)(chr12:g.12007967::chr15:g.88657925) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-002 6390-T01-I M6-185	P-0026 390-T0 1-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120168 63	5	155	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12016863::chr15:g.88651131) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-002 6286-T01-I M6-186	P-0026 286-T0 1-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120303 95	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12030395::chr15:g.88611476) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-002 5987-T01-I M6-187	P-0025 987-T0 1-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120289 42	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12028942::chr15:g.88613114) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 5507-T01-I M6-188	P-0025 507-T0 1-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120240 78	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12024078::chr15:g.88544573) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-002 3257-T01-I M6-189	P-0023 257-T0 1-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120149 98	5	155	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) fusion (ETV6 exons 1 to 4 fused to NTRK3 exons 14 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12014998::chr15:g.88662190) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-001 0562-T01-I M5-190	P-0010 562-T0 1-IM5		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120289 03	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 14-20): t(12;15)(p13.2;q25.1)(chr12:g.12028903::chr15:g.88581658) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-002 1974-T01-I M6-191	P-0021 974-T0 1-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120284 17	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 fused to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028417::chr15:g.88550500) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-000 2978-T02-I M5-192	P-0002 978-T0 2-IM5		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120285 76	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			NA Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-000 6480-T01-I M5-193	P-0006 480-T0 1-IM5		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120295 48	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12029548::chr15:g.88498145) Protein fusion: in frame (ETV6-NTRK3)	NA

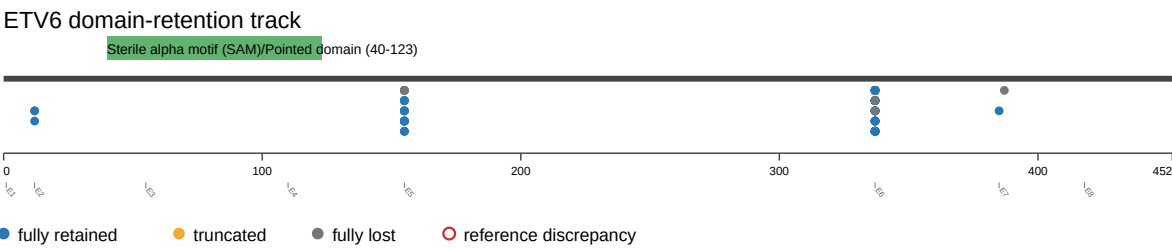
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 8060-T01-I M6-194	P-0018 060-T0 1-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120370 37	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12037037::chr15:g.88514531) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-001 6507-T04-I M6-195	P-0016 507-T0 4-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120237 09	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-001 6507-T03-I M6-196	P-0016 507-T0 3-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120237 09	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20: t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-000 2978-T07-I M6-197	P-0002 978-T0 7-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120285 76	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-001 6507-T01-I M6-198	P-0016 507-T0 1-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120237 09	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-000 4737-T01-I M5-199	P-0004 737-T0 1-IM5		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120068 09	4	155	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) - NTRK3 (NM_001012338) Fusion (ETV6 exons 1 to 4 fused to NTRK3 exons 13 to 20): t(12;15)(p13.2;q25.3)(chr12:g.12006809::chr15:g.88632786) Protein fusion: in frame {ETV6-NTRK3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 0668-T01-I M5-200	P-0010 668-T0 1-IM5		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120291 09	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12029109::chr15:g.88513282) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-001 4872-T03-I M6-201	P-0014 872-T0 3-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120289 46	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028946::chr15:g.88548029) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-001 4872-T02-I M6-202	P-0014 872-T0 2-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120289 46	5	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028946::chr15:g.88548029) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-001 3196-T04-I M6-203	P-0013 196-T0 4-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120360 72	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072::chr15:g.88501357) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-000 7772-T01-I M5-204	P-0007 772-T0 1-IM5		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120202 57	5	155	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257::chr15:g.88600844) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-003 3921-T03-I M6-205	P-0033 921-T0 3-IM6		NTRK 3-ETV 6	NTR K3	in-frame	five_prime	120311 61	6	337	True	retained	1.0	False	Sterile alpha motif (SAM)/ Pointed domain			ETV6 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534) Protein Fusion: in frame {ETV6:NTRK3}	NA

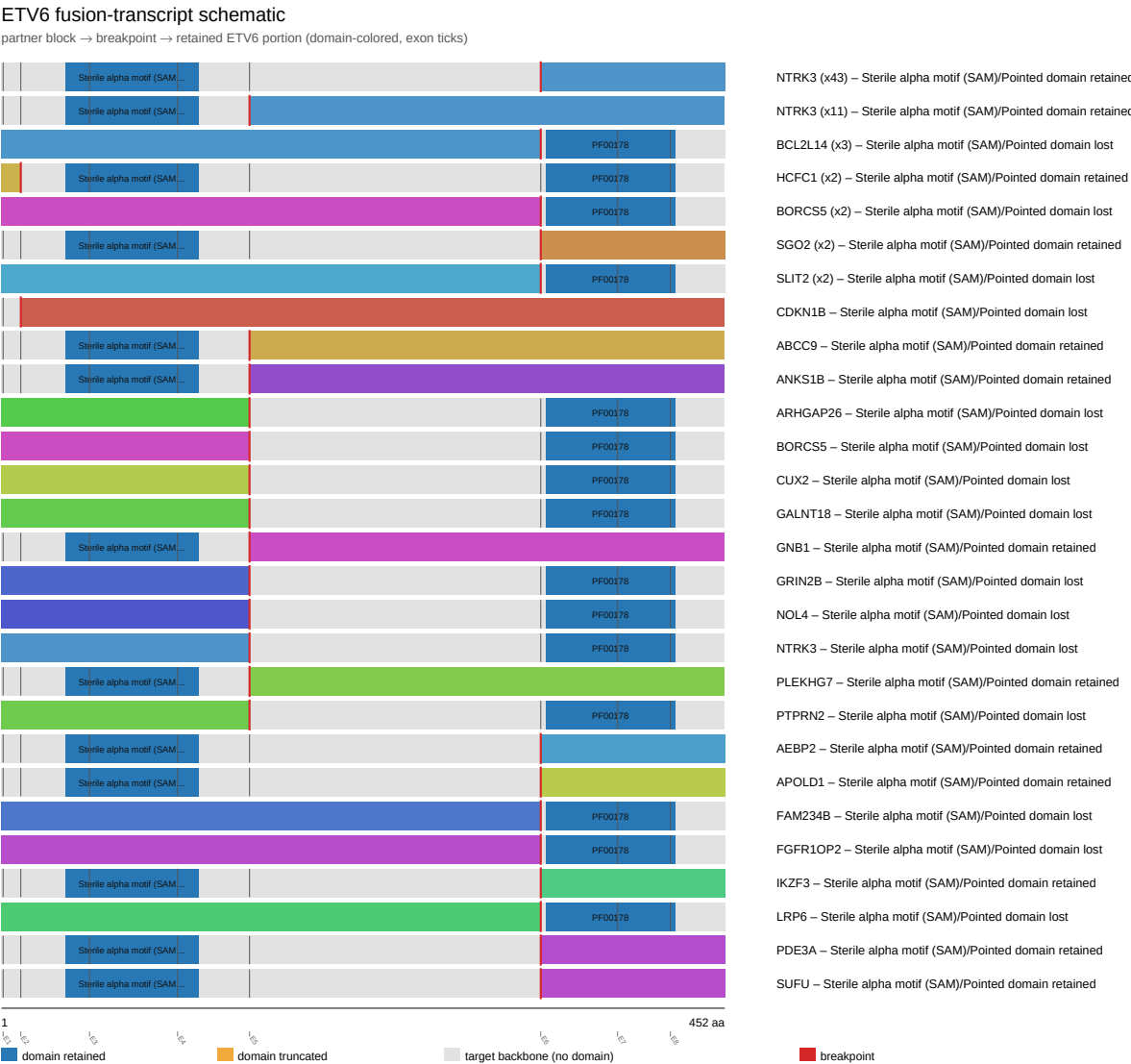
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 1048-T02-IM6-207	P-0011 048-T02-IM6		PPIL2-ETV6	PPI L2	unknown	three_prime	12038867	7	387	False	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		PPIL2 (NM_014337) - ETV6 (NM_001987) Rearrangement : t(12;22)(p13.2;q11.22)(chr12:g.12038867::chr22:g.22041881) Protein Fusion: mid-exon {PPIL2:ETV6}	NA
EVT-P-004 0504-T01-IM6-211	P-0040 504-T01-IM6		SLIT2-ETV6	SLIT2	out-of-frame	three_prime	12034059	6	337	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		SLIT2 (NM_004787) - ETV6 (NM_001987) rearrangement: t(4;12)(p15.31;p13.2)(chr4:g.20374490::chr12:g.12034059) Protein Fusion: out of frame {SLIT2:ETV6}	NA
EVT-P-004 0504-T02-IM6-212	P-0040 504-T02-IM6		SLIT2-ETV6	SLIT2	out-of-frame	three_prime	12034059	6	337	True	lost	0.0	False		Sterile alpha motif (SAM)/ Pointed domain		SLIT2 (NM_004787) - ETV6 (NM_001987) rearrangement: t(4;12)(p15.31;p13.2)(chr4:g.20374490::chr12:g.12034059) Protein Fusion: out of frame {SLIT2:ETV6}	NA

Figures

domain retention outliers



fusion schematic

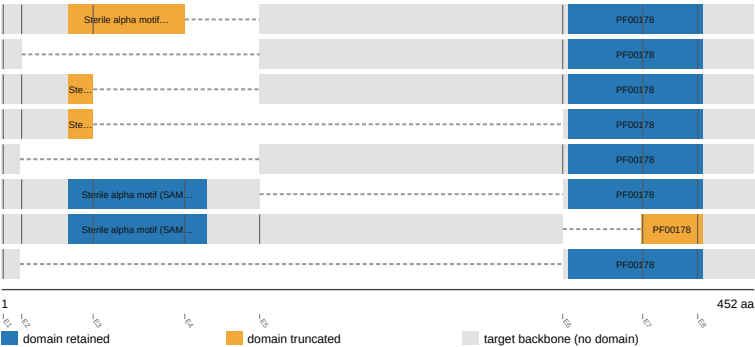


Showing the top 28 of 30 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic

ETV6 intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block



- 1-exon deletion (in-frame) (x5)
- 3-exon deletion (out-of-frame) (x4)
- 2-exon deletion (in-frame) (x4)
- 3-exon deletion (in-frame) (x3)
- 3-exon deletion (out-of-frame) (x2)
- 1-exon deletion (in-frame) (x2)
- 1-exon deletion (out-of-frame) (x2)
- 4-exon deletion (out-of-frame)

reference comparison

Reference vs msk_impact_50k_2026

